

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)

11
priors compared

31,623
held-out plate appearances

3.05%
held-out HR rate

0.6020
best AUC · KNN comps

served prior

Held-out scores

Fitted on plate appearances before 2026-08-29, scored on the 31,623 that came after (actual HR rate 3.05%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5862	1.61×	0.02950	0.13544	0.00535
KNN comps Prior is the pooled rate of the nearest comparables.	0.6020	1.55×	0.02945	0.13442	0.00478
SVR Support-vector regression on the same feature matrix.	0.5963	1.55×	0.02949	0.13515	0.00527
Random forest Forest regression; a learned similarity metric.	0.5956	1.47×	0.02948	0.13481	0.00643
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6020	1.55×	0.02945	0.13442	0.00478
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6020	1.55×	0.02945	0.13442	0.00478
Linear regression Ordinary least squares on the same features.	0.5963	1.51×	0.02947	0.13486	0.00416
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5974	1.55×	0.02949	0.13494	0.00556
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5964	1.53×	0.02948	0.13486	0.00505
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5962	1.55×	0.02949	0.13495	0.00593

BLEND WEIGHTS (FOREST VARIANT)

KNN 1.00 SVR 0.00 RF 0.00

SHRINKAGE K PER PRIOR

league 252 knn 252 svr 252 rf 252 core 252 ensemble 252
linear 252 logistic 252 nn 252 xgb 252

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,518 PA, actual HR rate 2.44%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5671	1.39×	0.02386	0.11499	0.01167
KNN comps Prior is the pooled rate of the nearest comparables.	0.5821	1.39×	0.02386	0.11479	0.01119
SVR Support-vector regression on the same feature matrix.	0.5696	1.28×	0.02403	0.11926	0.01804
Random forest Forest regression; a learned similarity metric.	0.5845	1.16×	0.02400	0.11611	0.01452
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5821	1.39×	0.02386	0.11479	0.01119
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5821	1.39×	0.02386	0.11479	0.01119
Linear regression Ordinary least squares on the same features.	0.5744	1.05×	0.02396	0.11712	0.01538
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5796	1.28×	0.02395	0.11633	0.01209
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5761	1.39×	0.02393	0.11629	0.01479
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5798	1.16×	0.02403	0.11727	0.01631

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Munetaka Murakami 18.2%, Miguel Vargas 15.8%, Kyle Teel 13.7%, Andrew Benintendi 13.5%, Jo Adell 13.4%, Colson Montgomery 12.2%
Flat league	5/6	Munetaka Murakami 16.6%, Miguel Vargas 14.2%, Kyle Teel 13.5%, Jo Adell 12.7%, Andrew Benintendi 12.1%, Patrick Bailey 11.6%
KNN comps	6/6	Munetaka Murakami 18.2%, Miguel Vargas 15.6%, Kyle Teel 14.2%, Jo Adell 13.2%, Andrew Benintendi 13.0%, Colson Montgomery 12.3%
SVR	5/6	Munetaka Murakami 18.7%, Miguel Vargas 15.6%, Jo Adell 13.7%, Kyle Teel 13.1%, Andrew Benintendi 12.5%, Patrick Bailey 11.8%
Random forest	6/6	Munetaka Murakami 18.4%, Miguel Vargas 15.3%, Kyle Teel 14.0%, Jo Adell 13.4%, Andrew Benintendi 12.7%, Colson Montgomery 12.3%
Core (KNN+SVR)	6/6	Munetaka Murakami 18.2%, Miguel Vargas 15.6%, Kyle Teel 14.2%, Jo Adell 13.2%, Andrew Benintendi 13.0%, Colson Montgomery 12.3%
Forest blend	6/6	Munetaka Murakami 18.2%, Miguel Vargas 15.6%, Kyle Teel 14.2%, Jo Adell 13.2%, Andrew Benintendi 13.0%, Colson Montgomery 12.3%
Linear regression	6/6	Munetaka Murakami 18.0%, Miguel Vargas 15.6%, Kyle Teel 13.8%, Jo Adell 13.3%, Andrew Benintendi 13.2%, Colson Montgomery 12.6%
Logistic regression	6/6	Munetaka Murakami 18.2%, Miguel Vargas 15.8%, Jo Adell 13.7%, Andrew Benintendi 13.4%, Colson Montgomery 13.1%, Kyle Teel 12.9%
Neural net (torch)	6/6	Munetaka Murakami 18.3%, Miguel Vargas 15.7%, Jo Adell 13.7%, Kyle Teel 13.3%, Andrew Benintendi 13.1%, Colson Montgomery 12.5%
XGBoost	6/6	Munetaka Murakami 18.6%, Miguel Vargas 15.6%, Kyle Teel 13.9%, Jo Adell 13.5%, Andrew Benintendi 12.8%, Colson Montgomery 12.4%

Hitter by hitter

The top 40 bats by the served probability.

HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (T
1Munetaka Murakami	slate18.2%	16.6%	18.2%	18.7%	18.4%	18.2%	18.2%	18.0%	18.2%	
2Miguel Vargas	CWS15.8%	14.2%	15.6%	15.6%	15.3%	15.6%	15.6%	15.6%	15.8%	
3Kyle Teel	CWS13.7%	13.5%	14.2%	13.1%	14.0%	14.2%	14.2%	13.8%	12.9%	
4Andrew Benintendi	INDIS13.5%	12.1%	13.0%	12.5%	12.7%	13.0%	13.0%	13.2%	13.4%	
5Jo Adell	LEslate13.4%	12.7%	13.2%	13.7%	13.4%	13.2%	13.2%	13.3%	13.7%	
6Colson Montgomery	INDIS12.2%	11.1%	12.3%	11.8%	12.3%	12.3%	12.3%	12.6%	13.1%	
7Patrick Bailey	CLE12.2%	11.6%	12.2%	11.8%	11.9%	12.2%	12.2%	12.0%	11.6%	
8Nathaniel Lowe	CLE11.3%	11.0%	11.3%	11.1%	11.6%	11.3%	11.3%	11.2%	11.1%	
9Chase DeLauter	CLE11.1%	11.2%	11.0%	10.9%	11.0%	11.0%	11.0%	10.9%	10.8%	
10Travis Bazzana	CLE11.0%	11.0%	10.7%	10.5%	11.0%	10.7%	10.7%	10.8%	10.5%	
11José Ramírez	CLE10.3%	10.2%	10.1%	9.9%	10.2%	10.1%	10.1%	10.4%	10.6%	
12Tristan Peters	CWS10.1%	10.7%	10.3%	10.6%	10.5%	10.3%	10.3%	10.4%	10.5%	
13Sam Antonacci	CWS9.4%	11.2%	9.7%	8.8%	9.8%	9.7%	9.7%	9.6%	9.4%	
14Chase Meidroth	CWS8.2%	9.5%	8.6%	8.9%	8.9%	8.6%	8.6%	8.7%	8.5%	
15Petey Halpin	CLE8.0%	9.2%	8.0%	8.3%	8.6%	8.0%	8.0%	9.2%	8.6%	
16Brayan Rocchio	CLE7.0%	7.5%	6.8%	7.1%	6.6%	6.8%	6.8%	6.7%	6.7%	
17Brenton Doyle	CWS6.6%	8.8%	7.4%	7.5%	6.9%	7.4%	7.4%	7.3%	7.1%	
18Steven Kwan	CLE4.8%	8.1%	5.8%	4.2%	4.2%	5.8%	5.8%	3.5%	5.0%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.